

PRODUCT SPECIFICATION • V1.0

Product Requirements Document (PRD)

Comprehensive Product Specification & Feature Blueprint

PROJECT

Gymholik SaaS

VERSION

1.0.0

AUTHOR

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DATE

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1. Executive Summary & Vision

Gymholik is an enterprise-grade, multi-tenant Software-as-a-Service (SaaS) platform built specifically to streamline the operational, financial, and client-management lifecycles of modern fitness clubs and gyms. Operating in a high-density, multi-user environment, gym businesses require strict multi-tenant data segregation, intuitive role-specific operational interfaces, automated dues tracking, high-speed attendance check-ins, and dedicated personal training workflows.

The system eliminates manual paper logbooks, uncollected receivables, unauthorized gym access, and fractured communication between gym owners, front-desk staff, personal trainers, and gym members.

Core Value Proposition

Provide gym owners with complete financial control and anti-theft guards (audit trails & payment locks), staff with frictionless member check-in & fee collection, trainers with client accountability and weight progress notes, and members with transparent digital access.

2. Target Personas & Stakeholder Analysis

Persona	Role Identifier	Primary Objectives & Responsibilities	Key Frustrations Solved
Gym Owner	OWNER	P&L oversight, revenue tracking, fee collection, staff permissions, plan creation, trainer approval.	Cash leakage, untracked overdue memberships, staff altering financial records, inability to view audit logs.
Front-Desk Staff	STAFF	Speedy check-in desk, searching members by phone/name, collecting dues, registering new admissions.	Slow paper check-ins, manual expiry calculation, complex software interfaces during morning/evening rushes.
Personal Trainer	TRAINER	Monitoring assigned client attendance, recording workout & weight progress notes, client profile management.	Losing track of client progress, lack of member assignment visibility, clunky communication.
Super Administrator	SUPER_ADMIN	Platform health, gym tenant provisioning, subscription licensing, global system monitoring.	Manual tenant database management, multi-tenant cross-contamination risks, subscription enforcement.

3. Functional Requirements & Feature Matrix

3.1 Multi-Tenant Gym Onboarding & Lifecycle

- **Trial Registration:** Prospective gym owners sign up with gym name, owner name, owner phone, and password. An instant 14-day free trial is activated.
- **Unique Gym Code Generation:** Each tenant receives a distinct, collision-free uppercase alphanumeric gym code (e.g., TITANF4067) used by staff and trainers during registration.
- **Trial Expiration Guards:** Upon trial expiry, write operations are suspended until subscription extension or payment verification by Super Admin.

3.2 Role-Based Access Control (RBAC) & Tab Visibility

Navigation and access are strictly partitioned to prevent privilege escalation and unauthorized financial disclosure:

Feature / Route	Owner	Staff	Trainer	Functional Guarantee
Financial Dashboard (Revenue, Outstanding)	Full Access	Full Access	Forbidden	Trainers never view gym turnover, daily collection, or dues metrics.
Member Directory	All Members	All Members	Assigned Only	Trainers only see active members officially assigned to their roster.
Attendance Check-In (Raised Center Tab)	Full Access	Full Access	Assigned Only	Quick check-in with expired membership warnings and sage visual pulse.
Payments & Dues Collection	Full Access	Collection Only	Forbidden	Staff can record collections; only Owner can execute payment reversals.
Client Progress Notes & Weight Chart	Member Detail	Member Detail	Dedicated Tab	Trainers have dedicated daily notes workspace with chart progression.
Gym Settings, Staff & Plans	Full Access	Forbidden	Forbidden	Staff and trainers cannot modify pricing, admission fees, or staff list.

3.3 Member Management & Subscription Engine

- **Member Registration:** Collects full name, phone number, gender, date of birth, emergency contact, photo, and initial membership plan.
- **Dynamic Plan Calculation:** Start date + plan duration days calculates expiry date. Admission fee + plan price equals total payable. Supports partial payment with balance dues.
- **Membership Status Engine:** Automatically derives status: ACTIVE (valid expiry), EXPIRED (expiry date in past), or DUE (outstanding balance).

3.4 Attendance & Access Control

- **Manual Quick Check-in:** Real-time debounced search by name, phone, or member code with single-tap check-in.
- **Expired Access Interception:** If an expired member attempts check-in, the system triggers audio/haptic warning and a modal prompt for renewal.
- **QR Code Scanning:** Instant camera scanner validating member ID against gym tenant.

3.5 Financial Ledger & Payment Reconciliation

- **Payment Modes:** Support for CASH, UPI, CREDIT/DEBIT CARD, and BANK TRANSFER.
- **Receipt Generation:** Automated branded PDF receipt generation with breakdown of admission fee, plan fees, GST, and balance.
- **Owner Reversals:** Accidental or fraudulent transactions can only be reversed by gym owners with mandatory audit reasons.

4. Non-Functional Requirements (NFRs)

Security & Tenant Isolation

Zero tenant cross-talk. Strict ThreadLocal context combined with MongoDB query injection ensures gym data is 100% isolated. Passwords hashed with BCrypt (strength 12).

Latency & Responsiveness

API response time < 150ms at p95. App tab switching instantaneous with client-side React Query caching and zero UI layout shifts.

Offline & Fault Resilience

Cached member directories for offline search. Background synchronization when network connection is restored.

Auditability & Compliance

Immutable audit logging for staff creations, payment reversals, plan alterations, and member status overrides.

5. Success Metrics & Key Performance Indicators (KPIs)

- **Check-in Speed:** Average time from front-desk member arrival to check-in confirmation < 3 seconds.
- **Fee Recovery Rate:** Reduce gym uncollected dues by > 35% within 60 days of platform adoption.
- **Platform Retention:** Monthly tenant renewal rate > 94% post-trial period.